Survival-supervised deconvolution identifies a reproducible tumor–stroma prognostic axis in pancreatic cancer

Amber M. Young[a], Alisa Yurovsky[b], Xianlu Laura Peng[c,d], Jen Jen Yeh[c,d,e], Didong Li[a], Naim U. Rashid[a,d,*]

[a]Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[b]Department of Biomedical Informatics, Stony Brook University, Stony Brook, NY 11794, USA

[c]Department of Pharmacology, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[d]Lineberger Comprehensive Cancer Center, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[e]Department of Surgery, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA

[*]Corresponding author: nur2@email.unc.edu

Abstract

Pancreatic ductal adenocarcinoma (PDAC) outcomes reflect coordinated tumor-intrinsic and stromal states, but bulk-transcriptomic deconvolution typically lacks survival supervision. We developed DeSurv, a survival-supervised matrix factorization framework that jointly learns gene programs and survival associations. In TCGA and CPTAC PDAC, DeSurv recovered a compact three-program structure organized along a co-varying tumor–stroma axis, with classical tumor/restCAF-like stroma at one pole and basal-like tumor/proCAF-like stroma at the high-risk pole. This axis reflects cross-sample covariation rather than demonstrated single-cell co-expression or spatial adjacency. Across five external cohorts, each standard-deviation shift toward the basal-like/proCAF pole was associated with worse survival (pooled HR, 1.50; 95% CI, 1.31–1.72), and dichotomized high-risk status was associated with HR 1.79 (95% CI, 1.41–2.28). The score remained independently prognostic after PurIST and DeCAF adjustment. DeSurv performed comparably to simpler survival-supervised predictors; its contribution is a reproducible, source-attributable tumor–stroma prognostic axis recoverable from bulk PDAC transcriptomes.

Pancreatic ductal adenocarcinoma (PDAC) is the predominant form of pancreatic cancer and remains among the deadliest solid tumors, with poor long-term survival and limited durable responses to existing therapies[1]. Bulk transcriptomic analyses have identified reproducible malignant programs, most consistently basal-like and classical tumor states[2–4], along with stromal programs that include tumor-promoting proCAF and tumor-restraining restCAF states[5]. Single-cell and spatial profiling further show that PDAC lesions contain diverse malignant, stromal, immune, and other microenvironmental compartments that can adopt coordinated transcriptional states within the same tumor[6–8]. Whether joint configurations of malignant and microenvironmental programs provide prognostic information beyond discrete tumor or stromal labels remains incompletely understood.

Addressing this question at cohort scale still largely depends on bulk RNA profiling, a scalable platform for prognostic modeling[9,10] in which each profile reflects a composite signal from malignant, fibroblast, immune, and other cellular sources[11]. Nonnegative matrix factorization (NMF) and related deconvolution approaches can decompose these profiles into additive, interpretable gene programs[3,12–14]. In PDAC, such approaches have yielded transcriptomic molecular subtypes[2,4,15], virtual microdissection of basal-like and classical malignant programs[3], and compartment-specific deconvolution of tumor and stromal signals[5,14].

These successes also expose a limitation of a common workflow: gene programs are often discovered without reference to outcome and evaluated for clinical association only after factorization[2,15]. Because unsupervised NMF optimizes reconstruction rather than prediction, its factors may capture dominant expression structure driven by tissue composition or other outcome-neutral variation rather than prognostic biology[9,16]. In PDAC, exocrine, stromal, and other low-purity signals can confound unsupervised subtype discovery and may be less consistently prognostic than tumor-intrinsic basal-like and classical programs[4,5,17], part of a broader pattern in which tumor purity confounds transcriptomic correlation and clustering across cancer types[18]. Reconstruction alone also leaves the NMF basis generally non-identifiable without additional assumptions or constraints[19–21], potentially contributing to cross-study variability among unsupervised cancer subtype taxonomies[17,22].

These limitations raise a central question: which bulk-transcriptomic programs carry reproducible prognostic information in PDAC, and how are they distributed across malignant, stromal, immune, and other tumor-microenvironmental compartments? Neither reconstruction-driven nor outcome-driven analysis is sufficient on its own. Unsupervised factorization can overrepresent composition-driven variation, whereas survival-based gene selection can identify outcome-associated features but does not, by itself, resolve their cellular origin.

Here, we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox proportional hazards modeling, allowing gene programs to be shaped during discovery by both reconstruction fidelity and survival association. Supervision is applied to the gene-program matrix while preserving the interpretation of sample loadings as mixture coefficients, and trained programs can score new samples by projection without retraining. We evaluate DeSurv using two tests designed to avoid circularity: recovery of predefined latent programs in simulations where the true gene sets are known a priori, and out-of-sample generalization across five independent PDAC cohorts in which no model parameters were estimated.

In simulations, DeSurv recovers prognostic programs more reliably than standard NMF, with the largest gains when high-variance and prognostic subspaces are distinct and no measurable advantage over standard NMF when survival signal is absent. In PDAC, DeSurv identifies a compact three-program structure organized around a co-varying tumor–stroma axis, reflecting coordinated variation in tumor and stromal gene programs across patients rather than single-cell co-expression or spatial adjacency. At one pole, classical tumor identity co-varies with restCAF-like stroma, whereas the high-risk pole is marked by basal-like tumor identity and proCAF-like stroma. The resulting score generalizes across five independent PDAC cohorts without retraining and remains prognostic beyond established PurIST and DeCAF classifications. Together, these findings identify a reproducible tumor–stroma prognostic axis in PDAC that can be recovered de novo from bulk transcriptomes using survival-supervised matrix factorization.

Results

DeSurv learns survival-associated gene programs from bulk transcriptomes

To identify prognostic yet interpretable gene programs, we applied DeSurv (Fig. 1A, Methods), which shapes programs by both expression reconstruction and survival association acting through the gene-program matrix, preserving the interpretation of sample loadings as mixture coefficients, so that trained programs can be projected into new cohorts without retraining.

We applied DeSurv to the PDAC training cohorts from TCGA and CPTAC[23,24] (Methods). Cross-validated concordance plateaued from $k = 3$ through $k = 12$ (Supplementary Fig. S4), and Bayesian optimization over rank and supervision strength with a one-standard-error rule selected a parsimonious three-factor solution, $k = 3$ at $\alpha = 0.334$ (Fig. 1B). The peak cross-validated C-index was 0.655 across 273 patients with 139 events. By contrast, standard unsupervised NMF diagnostics, including reconstruction residuals, cophenetic correlation and silhouette width[13,25], did not identify a consistent rank in PDAC (Supplementary Fig. S5). All rank and hyperparameter selection was performed within the training set before projection into validation cohorts.

Survival supervision reshapes PDAC factor structure

To determine how survival supervision shaped the recovered factor structure, we fit standard NMF at the same rank selected for DeSurv ($k = 3$) and compared each method's factor-specific gene rankings with established PDAC gene programs (Fig. 2A–B). Matching k across methods controls for differences in model complexity when comparing the effect of survival supervision on factor content; a sensitivity analysis using NMF's independently selected ranks (elbow $k = 5$, BO $k = 7$) is provided in Supplementary Table S5. We compared the biological content of each factor and then quantified how much each contributed to reconstruction and survival.

We refer to the three DeSurv factors as D1–D3 and the three NMF factors as N1–N3. In the DeSurv factorization, D1 correlated with classical tumor programs and restCAF-like stromal signatures, D2 with proCAF-like programs, and D3 with basal-like tumor programs (Fig. 2A). In the DeSurv solution, the dominant survival-associated signal is aligned with a tumor–stroma axis: higher D1 scores were associated with a classical/restCAF-like, lower-risk state, whereas higher basal-like tumor and proCAF-like stromal scores were associated with higher risk. This organization is consistent with prior evidence that joint tumor-intrinsic and CAF state stratification improves prognostic resolution in PDAC[5]. No DeSurv factor showed positive correlation with exocrine-associated programs above the display threshold, including DECODER Exocrine, Collisson Exocrine and Bailey aberrantly differentiated endocrine–exocrine (ADEX) programs (Fig. 2A).

Standard NMF produced a different factor organization (Fig. 2B). One factor, N2, was strongly correlated with exocrine-associated programs and negatively correlated with immune and stromal signatures. A second factor, N1, correlated with malignant-epithelial identity and carried both classical and basal-like markers. A third factor, N3, correlated with immune, fibroblast and extracellular matrix (ECM)-related programs, including Puleo Desmoplastic, SCISSORS proCAF and restCAF[26], Maurer Immune and ECM[27], and DECODER Activated Stroma. Thus, at $k = 3$, standard NMF devoted one factor to exocrine-associated variation and aggregated tumor-intrinsic and microenvironmental programs into two broad factors, whereas DeSurv separated basal-like tumor, proCAF-like stroma and a classical/restCAF-like survival-aligned program.

NMF seeks a low-rank *reconstruction* of the transcriptome; many factorizations can reconstruct comparably well, and the unsupervised objective has no incentive to prefer prognostically informative directions. We therefore quantified each factor's contribution using two metrics (Fig. 2C; Supplementary Information, Sections 10–11): a normalized reconstruction Shapley value, which partitions reconstruction contribution across factors and sums to 100%, and survival contribution, defined as the change in Cox partial log-likelihood ($\Delta\ell$) when that factor is dropped from a k-factor Cox model. Reconstruction and survival contribution were decoupled: all three NMF factors had small $\Delta\ell$ despite spanning a wide range of reconstruction contribution, whereas in the training set (pooled TCGA + CPTAC) DeSurv concentrated nearly all survival contribution into D1 ($\Delta\ell = 66.4$ from only 18.8% of reconstruction; per-factor values in Fig. 2C). Thus, in PDAC, factors that best reconstruct expression are not necessarily the factors that best explain survival, consistent with tumor-purity analyses across cancer types[18].

To further characterize the relationship between the two factorizations, we examined pairwise W-matrix gene-loading correlations across all 1,970 genes (Fig. 2D; Spearman correlations over full loading profiles, since top-gene subsets can overstate preservation). At $k = 3$, NMF captured the malignant-epithelial compartment largely in a single factor, N1, which carried both classical and basal-like markers and therefore is labeled *tumor* rather than classical. N1 mapped strongly onto DeSurv's basal-like tumor factor, D3 (Spearman $\rho = 0.84$), while NMF's microenvironmental factor, N3, mapped onto DeSurv's proCAF-like stromal factor, D2 ($\rho = 0.87$). In contrast, DeSurv's most prognostic factor, D1, was only weakly correlated with all three NMF factors (all $\rho \leq 0.36$). Because D1 couples classical tumor and restCAF-like stromal signatures into a single survival-aligned program, it is not simply a reweighting of an NMF factor: little of its gene-loading profile is linearly reconstructable from the matched-rank NMF basis (Supplementary Table S2). NMF's exocrine factor, N2, had no comparably strong DeSurv counterpart (maximum $\rho = 0.61$), consistent with reduced emphasis on exocrine-associated variation under survival supervision. Together, these analyses indicate that, at $k = 3$, standard NMF separated broad tumor, stromal/immune and exocrine-associated structure, whereas DeSurv produced a survival-aligned tumor–stroma program not closely matched by any single NMF factor.

Survival-aligned programs generalize across independent PDAC cohorts

We next tested whether the programs learned in the TCGA and CPTAC training cohort transferred to five independent PDAC cohorts without retraining (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq and Puleo; non-metastatic, treatment-naive specimens; Supplementary Table S1), projecting the training-derived programs into each validation expression matrix ($Z = \widetilde{W}^\top X_{\text{new}}$) to obtain a continuous per-factor score for each sample.

Across the 5 validation cohorts ($n = 616$), the DeSurv linear predictor (oriented so that higher values indicate higher risk) was consistently associated with survival in a stratified Cox model (pooled HR per SD = 1.50, 95% CI 1.31–1.72, $P = 2.6 \times 10^{-9}$; Fig. 3A). At the matched rank of $k = 3$, the analogous standard NMF predictor showed a weaker pooled association (HR per SD = 1.11, 95% CI 1.00–1.23, $P = 0.057$). The combined DeSurv linear predictor remained prognostic after simultaneous adjustment for PurIST and DeCAF classifications (per-SD HR = 1.33, 95% CI 1.13–1.56, $P < 0.001$; Table 1), indicating that the tumor–stroma axis captures information beyond established malignant and stromal subtype labels.

The continuous, stratified-Cox associations above are our primary analysis; to illustrate them clinically, we dichotomized validation samples into high- and low-risk groups using method-specific cutpoints independently optimized by cross-validation in the training data. The DeSurv linear predictor separated survival trajectories in the pooled validation cohort (HR = 1.79, 95% CI 1.41–2.28, $P < 0.001$; Fig. 3B), whereas standard NMF at $k = 3$ yielded weaker stratification (HR = 1.38, 95% CI 0.92–2.09, $P = 0.122$; Fig. 3C). The DeSurv high-risk group was enriched for both established poor-prognosis subtypes, basal-like by PurIST and proCAF by DeCAF (Fisher's exact test; Supplementary Fig. S7), whereas NMF risk groups at $k = 3$ showed basal-like enrichment but no significant proCAF enrichment (Supplementary Fig. S7), consistent with NMF not isolating the tumor–stroma axis.

Higher-rank NMF models achieved similar validation concordance in some cohorts but required more than twice as many factors and did not recover the same compact tumor–stroma axis: NMF at elbow-selected $k = 5$ and Bayesian-optimization-selected $k = 7$ improved over $k = 3$ (Supplementary Table S5), yet NMF's own rank-selection diagnostics did not converge on these ranks (Supplementary Fig. S5), and DeSurv's three-factor model was preferred by BIC over NMF's seven-factor model in three of five cohorts (Supplementary Table S6). Prediction alone likewise did not distinguish DeSurv from simpler survival-supervised models: its pooled transfer concordance (C-index = 0.62) was on par with penalized (lasso) Cox (0.61), Cox partial least squares (0.63) and supervised principal components (0.64), whereas unsupervised directions transferred poorly (standard NMF at $\alpha = 0$, 0.55; the leading principal component, 0.49), consistent with evidence that learned representations do not reliably out-predict penalized regression for out-of-sample prognosis[28]. DeSurv's contribution is therefore not higher predictive accuracy but the decomposition of a transferable prognostic signal into a compact, interpretable, source-attributable tumor–stroma program.

Table 1: Pooled external validation hazard ratios (per SD of the linear predictor) for DeSurv at $k = 3$ and standard NMF at BO-selected $k = 7$ ($\alpha = 0$), across the five validation cohorts ($n = 616$ patients, 414 events). Adjusted models include the PurIST tumor-intrinsic and DeCAF stromal molecular classifiers as covariates, with stratification by dataset. Both methods retain significant prognostic value after adjustment.

Method	Unadjusted HR per SD (95% CI), P	Adjusted HR per SD (95% CI), P
DeSurv ($k = 3$)	1.50 (1.31–1.72), < 0.001	1.33 (1.13–1.56), < 0.001
Std NMF ($k = 7$, BO $\alpha = 0$)	1.48 (1.34–1.64), < 0.001	1.47 (1.27–1.70), < 0.001

The tumor–stroma axis adds prognostic information beyond established PDAC classifiers

We next asked whether the D1 axis provided information beyond established discrete subtype calls. Across the pooled validation cohorts ($n = 616$, 414 events), binary PurIST and DeCAF classifications together explained 28% of the within-cohort variance in the continuous D1 score, so D1 aligned with known tumor-intrinsic and stromal subtypes while capturing a finer-grained continuum of tumor–stroma state.

D1 remained independently associated with survival after simultaneous adjustment for PurIST and DeCAF classifications (adjusted HR per SD = 0.82, 95% CI 0.74–0.92, $P = 0.0006$). Adding D1 to a stratified Cox model containing PurIST and DeCAF improved model fit (partial likelihood-ratio $\chi^2 = 11.6$ on 1 df, $P = 0.0006$) and increased concordance from 0.593 to 0.621. The tumor–stroma axis therefore captures prognostic variation not fully represented by either classifier alone.

This axis was not specific to the DeSurv optimizer. Three independent survival-supervised approaches (supervised principal components, Cox partial least squares and sparse Cox) recovered scores highly correlated with D1 in the training cohort (mean $|r| = 0.83$), whereas the leading unsupervised direction did not ($|r| = 0.10$; Supplementary Information, supervised-method recovery). These results indicate that D1 reflects a reproducible survival-associated expression direction rather than an idiosyncrasy of a single factorization algorithm.

Finally, in the fully annotated external PACA-AU cohorts, the DeSurv score remained prognostic after adjustment for stage, grade, age and sex, and further adjustment for tumor purity did not eliminate the association in the two PACA-AU cohorts with purity annotations (Supplementary Table S4). In the smaller Dijk and Moffitt cohorts, the DeSurv score was not significantly associated with survival before adjustment, and the Moffitt purity-adjusted analysis was underpowered ($n = 29$). These analyses support the tumor–stroma axis as a prognostic continuum that is not reducible to established subtype labels, standard clinical covariates or tumor purity.

In an exploratory, hypothesis-generating analysis, we also projected the trained programs into three bulk treated PDAC cohorts (the metastatic Rash-Accept series and the borderline-resectable/locally advanced Linehan cohort); because treatment, stage and cohort were confounded, these associations are prognostic, not predictive. The proCAF program D_2 showed a consistent protective direction on overall survival (reaching significance in the less-advanced Linehan cohort), whereas the basal–classical axis was setting-dependent, attenuating in metastatic disease as it does for established classifiers (Supplementary Information, Section 20).

DeSurv recovers prognostic gene programs when variance and prognosis diverge

We asked whether survival supervision improves recovery of prognostic gene programs when prognostic variation is not the dominant source of transcriptomic variance, a regime that probes weak prognostic signal rather than all transcriptomic settings. We designed simulations with known ground-truth latent structure, in which prognostic gene programs explained low variance relative to outcome-neutral background signals and survival times were generated from latent factor scores via a Cox model ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Materials and Methods, Simulation Studies). We compared DeSurv to standard NMF ($\alpha = 0$), where factors are learned by reconstruction error alone and survival associations are estimated by a separately fit Cox model; both were tuned via BO over cross-validated C-index, so differences reflect the role of supervision during factorization, not during model selection.

In the prognostic scenario, where prognostic programs explained low variance relative to outcome-neutral programs, DeSurv consistently achieved higher C-index (Fig. 4A) and substantially higher precision for recovering the true prognostic gene programs (Fig. 4B). Here precision measures how well the learned factors recover the true prognostic programs, computed per program against its best-matching factor and averaged (Supplementary Information, Sections 6 and 9). The unsupervised baseline exhibited near-zero precision, indicating that variance-driven factorization failed to concentrate on the genes that actually drove survival. This indicates that outcome supervision during factorization improves recovery of prognostic programs that do not dominate expression variance. Across 100 replicates, DeSurv achieved median C-index 0.837 versus 0.724 for standard NMF ($\Delta = 0.113$, paired Wilcoxon $P < 0.001$). The gain in gene-program precision exceeded the gain in C-index, indicating that unsupervised factorization can achieve similar predictive performance while recovering different gene programs. Survival supervision therefore improves not only prediction but also recovery of the underlying prognostic biology.

To verify that these gains reflect genuine signal recovery rather than overfitting, we repeated the analysis under a null scenario ($\beta = 0$) and a mixed scenario with partial overlap between variance-dominant and prognostic structure (Supplementary Fig. S3). The benefit of survival supervision scaled with the separation between prognostic and variance-dominant structure: largest in the prognostic scenario ($\Delta = 0.113$), attenuated when variance and prognosis partially overlapped ($\Delta = 0.069$, paired Wilcoxon $P < 0.001$), and absent when no survival signal existed, where both methods yielded C-index near 0.5. DeSurv also recovered the true rank ($k = 3$) more frequently than standard NMF, which systematically under-selected near $k = 2$ (Fig. 4C).

Discussion

Survival-supervised factorization revealed that the dominant prognostic transcriptional structure in PDAC is a tumor–stroma axis, coupling classical tumor identity and restCAF-like stroma at one pole with basal-like tumor identity and proCAF-like stroma at the other. This axis generalized in a pooled analysis of five independent PDAC cohorts without retraining and remained prognostic after adjustment for PurIST and DeCAF, indicating that outcome is associated with the coordinated configuration of tumor-intrinsic lineage state and stromal composition, not fully captured by the compartment-specific classifiers evaluated here. Although purely supervised predictors fit directly to bulk expression achieved similar external discrimination, they returned a single, compartment-ambiguous risk score. DeSurv instead decomposed this prognostic signal into a small number of interpretable, source-attributable programs by coupling NMF to a Cox objective during program discovery.

This distinction is methodological as well as biological. In DeSurv, survival supervision acts on the transferable gene-program basis, W, while preserving the interpretation of sample loadings, H, as mixture coefficients, so the learned programs project into new samples without re-estimating the factorization. Prior survival-supervised NMF approaches have similarly integrated Cox objectives with matrix factorization, but generally supervise the latent sample representation or impose survival constraints on the factorization as a whole[29,30]. Deep survival-representation methods, including interpretable autoencoder- and attention-based Cox models, can learn prognostic latent features but typically return representations whose biological attribution depends on model-specific decoding or explanation steps[31,32]. DeSurv's contribution is to keep the prognostic objective tied to a fixed, source-attributable gene-program basis, making the recovered programs both interpretable and directly transportable across cohorts.

Biologically, DeSurv aligns the tumor–stroma axis with a single survival-associated program: the factor with the largest survival contribution, D1, links classical tumor and restCAF-like stromal genes, recovered de novo without predefined signatures while their survival associations are estimated during factorization. This organization is concordant with virtual microdissection[3], experimental microdissection[27], and unsupervised deconvolution[14], and with evidence that PDAC outcome is shaped by tumor-intrinsic state together with fibroblast and stromal programs[5,33–35] whose cellular states are themselves heterogeneous[36]. By contrast, standard NMF at the matched rank of $k = 3$ devoted one factor to exocrine-associated, low-purity variation[4], whereas DeSurv separated the microenvironmental programs at the selected rank.

This parsimony reflects both survival supervision and the model-selection framework. Survival supervision produced a broad concordance plateau across ranks, letting the one-standard-error rule select a compact three-factor model that standard NMF matched only at higher rank and greater complexity (Supplementary Table S5); standard NMF's own unsupervised rank-selection heuristics did not identify this solution (Supplementary Fig. S5). Joint optimization over rank and supervision strength, coupled to the one-standard-error rule, is therefore part of DeSurv's methodological contribution.

Our simulations clarify when this tradeoff is most useful. DeSurv's advantage was greatest when prognostic programs explained modest variance relative to outcome-neutral signals, and was absent under survival-null conditions. DeSurv is therefore not intended to replace unsupervised NMF in all settings. For exploratory analyses without clinical endpoints, settings with sparse or unreliable survival data, or studies focused on comprehensive biological characterization, unsupervised methods remain appropriate. The supervision weight, selected by Bayesian optimization, allows the data to determine the balance between reconstruction and survival association; at $\alpha = 0$, DeSurv reduces to standard NMF.

The same scope clarifies what DeSurv is not. It is not proposed as a competing molecular classifier: the relevant question is whether its factors add prognostic information beyond established classifiers, which they did after adjustment for PurIST and DeCAF (adjusted HR = 1.33, 95% CI 1.13–1.56, $P < 0.001$; Table 1). DeSurv is also distinct from reference-based deconvolution methods such as CIBERSORTx[37] and BayesPrism[38], which estimate cellular composition from single-cell or sorted-cell references; DeSurv instead learns prognostic latent gene programs directly from bulk expression without a cellular reference, and could be integrated with deconvolution-derived cell fractions to refine compartmental attribution.

Several limitations define the scope of these findings. DeSurv assumes proportional hazards, which may not hold under delayed or non-proportional effects, and the convergence guarantee applies to an idealized algorithm (Supplementary Theorem 1). Extensions to more flexible survival objectives are therefore a natural direction. The method was also applied here only to PDAC. Thus, although our simulations show that survival supervision can recover prognostic programs when variance-dominant and outcome-relevant subspaces diverge, the generality of this principle remains to be established across cancer types. Such studies will require multi-cancer benchmarking, including comparisons with strong penalized-regression baselines, which learned representations do not consistently outperform for out-of-sample prognosis[28].

The training and primary validation cohorts were non-metastatic and treatment-naive, so DeSurv was learned primarily in resectable disease. Projection of the trained programs into small treated cohorts ($n = 66$–85) suggested a consistent protective direction for the proCAF-like program and a more setting-dependent basal–classical axis, consistent with emerging evidence that PDAC stroma can have both tumor-promoting and tumor-restraining roles[39–41]. However, treatment, stage and cohort were confounded in these analyses, and the treated cohorts were small. We therefore regard these observations as exploratory and hypothesis-generating. Whether the proCAF-like program identifies patients who benefit from stroma- or myeloid-directed therapy will require prospective testing.

Finally, the tumor–stroma axis is defined from bulk gene programs: it reflects cross-patient covariation of classical-tumor and restCAF-like stromal signatures, not per-cell co-expression or spatial adjacency. Single-cell and spatial assays are better suited to resolve these relationships at cellular resolution. The bulk association is nevertheless supported by several orthogonal observations: D1 remained prognostic after PurIST and DeCAF adjustment, was recovered by independent survival-supervised methods but not by the leading unsupervised direction, and its classical-tumor identity was corroborated by GATA6 RNA in situ hybridization. Where tumor-purity annotations were available, purity adjustment further argued against the axis being driven solely by tumor purity. Adequately powered single-cell and spatial profiling of the same clinical settings is therefore an important next validation step, and mechanistic experiments will be needed to determine whether these tumor–stroma configurations causally shape PDAC progression.

More broadly, the conventional workflow of unsupervised discovery followed by retrospective clinical evaluation can lose prognostic structure when the dominant sources of expression variation are not those most relevant to outcome, as is common in bulk tumors where purity and tissue composition strongly shape transcriptomic profiles[18]; aligning factorization with the clinical question from the outset helps avoid this mismatch. Survival supervision also constrains the gene-program basis, which reconstruction alone leaves underdetermined and which may contribute to cross-study variability among unsupervised PDAC taxonomies[2,3,15]. Thus, survival supervision does more than improve prediction: it resolves the prognostic signal into reproducible, source-attributable gene programs, a property that distinguishes DeSurv from a conventional supervised risk score and should become increasingly useful as clinically annotated transcriptomic cohorts grow. Together, these findings support a model in which PDAC prognosis is associated with coordinated malignant–stromal transcriptional states that outcome-supervised deconvolution can recover from bulk tumors.

Methods

Problem formulation and notation

Let $X \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p \times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k \times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0, 1\}^n$ the event indicators ($\delta_i = 1$ for an observed event, $\delta_i = 0$ for censoring), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k \times n}$: for sample i with projected score $z_i = W^\top x_i$, the hazard is $h(t \mid x_i) = h_0(t) \exp(z_i^\top \beta)$, where h_0 is the baseline hazard. Defining factor scores by projection onto the learned gene programs ($Z = W^\top X$) rather than by the inferred loadings H lets the same basis score new cohorts without refitting, the property we exploit for external validation.

The DeSurv Model

DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with survival outcomes. The method operates in a survival-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0, 1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W, H, \beta) = (1 - \alpha)\, \mathcal{L}_{\mathrm{NMF}}(W, H) - \alpha\, \mathcal{L}_{\mathrm{Cox}}(W, \beta), \tag{1}$$

where $\mathcal{L}_{\mathrm{NMF}}(W, H)$ is the NMF reconstruction error (including an ℓ_2 penalty on H) and $\mathcal{L}_{\mathrm{Cox}}(W, \beta)$ is the elastic-net penalized partial log-likelihood. A key design choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients[12,42]. When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF, and β is fit post hoc by penalized Coxnet on the projected scores $Z = W^\top X$, used only for scoring and cross-validation without affecting the factorization. The Cox component also supports optional adjustment for sample-level covariates (for example, stage or grade); the analyses reported here use factor scores alone.

Optimization alternates updates for H, W, and β (Supplementary Information, Section 2). The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ). Although the joint problem is non-convex, the idealized alternating scheme converges to a stationary point under mild regularity conditions, and the implementation includes practical modifications (Supplementary Theorem 1). Closed-form update equations, the convergence proof, and BO control parameters are given in Supplementary Information, Sections 1–3.

Hyperparameter selection and cross-validation

Hyperparameters $(k, \alpha, \lambda, \xi, n_{\text{top}})$ were selected by maximizing the cross-validated Harrell C-index using Bayesian optimization (BO)[43,44], which finds near-optimal configurations using far fewer objective evaluations than grid or random search over mixed integer–continuous hyperparameter spaces, with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lies within one standard error of the maximum). All model selection was performed entirely within the training data using cross-validation; no validation cohort data were used during any stage of tuning, and unbiased generalization was assessed only on the external validation cohorts. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (Supplementary Information, Section 6). Before projection, each learned gene program (column of W) was restricted to its BO-selected number of top genes (denoted $\widetilde{W}$) and column-normalized (ℓ_2-norm scaling) by the same procedure used during training (Supplementary Information, Section 6); the resulting basis $\widetilde{W}$ was held fixed for all validation cohorts. In PDAC, BO selected $n_{\text{top}} = 270$ top genes per factor. The ridge penalties on the loadings (H) and gene programs (W) were held fixed at 0 in all analyses and were not tuned; factor scaling is instead controlled directly by column-normalization of W during optimization.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance (cross-validated C-index) rather than the unsupervised diagnostics commonly used for NMF rank selection[13,25]. In the PDAC training data, standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks, a known limitation of unsupervised rank-selection criteria that can disagree on the same dataset (Supplementary Fig. S5). Cross-validated concordance varied across candidate ranks but plateaued from $k = 3$ through $k = 12$ (Supplementary Fig. S4). The one-standard-error rule applied to joint BO over rank and supervision strength accordingly selected the parsimonious $k = 3$ at $\alpha = 0.334$.

External validation cohorts were scored by projecting new datasets onto the learned programs via $Z = \widetilde{W}^\top X_{\text{new}}$, and survival associations were evaluated using Harrell's C-index, hazard ratios, and log-rank statistics; because Harrell's C-index is rank-based and assesses discrimination only, not calibration or time-dependent prediction error, we additionally report a proper score (the integrated Brier score and its index of prediction accuracy relative to a Kaplan–Meier reference) as a sensitivity check (Supplementary Information, Proper-scoring sensitivity). Because the survival-supervised information lives in W rather than H, scoring is a pure matrix projection requiring no retraining or validation survival information. Related supervised-NMF methods that instead route survival supervision through H require sample-loading re-estimation on each new cohort[29,30]. Validation survival outcomes were therefore used only for downstream performance evaluation, not for model training, scoring, or hyperparameter tuning. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. Further training and validation details appear in Supplementary Information, Part I.

Simulation studies

Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\text{Gamma}(\text{shape} = 3, \text{rate} = 0.8)$ and cross-loadings onto other factors from $\text{Gamma}(1, 20)$; background gene loadings from $\text{Gamma}(2, 1)$ across all factors; and noise gene loadings from $\text{Gamma}(1, 20)$. Sample-level factor activities H were drawn from $\text{Gamma}(2, 2)$. Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the prognostic scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T W_{\text{true}}$, where W_{true} retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribution. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates. We considered three simulation scenarios: a prognostic scenario in which prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

Real-world datasets

We analyzed publicly available and controlled-access RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Eligible samples were primary PDAC tumors with a measured bulk expression profile and non-missing overall-survival time and event status; accrual and follow-up periods are as defined in each cohort's original publication (referenced in Supplementary Information, Section 7). Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\mathrm{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort separately, genes were ranked by mean expression and by variance independently; the top 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable; the intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. No missing-data imputation was performed: all analyses used complete cases, and analyses adjusted for the PurIST and DeCAF classifiers were restricted to samples with available classifier calls. Of the seven PDAC cohorts we considered, two were used for training (TCGA and CPTAC) and five were used for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo). To harmonize differences in scale across cohorts, filtered gene expression data were within-sample rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility); ranking removes platform- and library-size differences in expression scale that would otherwise dominate cross-cohort projection. More details about the datasets can be found in Supplementary Information, Section 7.

Per-cohort participant flow is summarized in Supplementary Table S1. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, missing tumor grade, or quality-control failure, yielding 273 analytic training samples (TCGA-PAAD 144, CPTAC-3 129; 139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation samples (414 events) across five cohorts: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181). These analytic samples provided ample events per estimated parameter: 139 training events for the 3-factor model (approximately 46 events per factor, well above the conventional minimum of ten) and 414 events across the external validation cohorts.

Data Availability

The training and validation datasets used in this study are publicly available or accessible through controlled-access repositories, and are de-identified; no Institutional Review Board (IRB) approval was required for their analysis. Training data were obtained from the Genomic Data Commons: TCGA-PAAD[45] and CPTAC-3[46]. Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830[47]; Puleo, E-MTAB-6134[48]), the Gene Expression Omnibus (GEO; Moffitt, GSE71729[49]), and the International Cancer Genome Consortium (ICGC): the PACA-AU array cohort is publicly available from the Gene Expression Omnibus (GSE36924), and the PACA-AU RNA-seq cohort from the European Genome–phenome Archive (EGA) study EGAS00001000154 under controlled access[50]. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study[5]; cohort details and sample sizes after quality control are summarized in Supplementary Information, Section 7. The treated bulk-tissue cohorts used in the exploratory translational analysis (Rash-Accept and the Linehan FOLFIRINOX$_{\pm}$CCR2-inhibitor series), and the COMPASS cohort used for the GATA6 RNA in situ hybridization comparison, are restricted-access clinical-trial datasets that are not publicly deposited; raw data are available from the respective study investigators under a data-use agreement, subject to the governing trial consents and institutional approvals. To preserve reproducibility without redistributing restricted data, the derived per-cohort summary statistics reported in the main text and Supplementary Information are provided in the reproducibility repository (file `results/treated_cohort_stats.rds`), so that every treated-cohort value can be inspected and verified.

Code Availability

All analyses were performed in R (version 4.6.0; R Core Team) using DeSurv (v1.0.1; this work), `NMF`[42], `glmnet`[51], `survival`, `ggplot2`, and `cowplot`; Bayesian optimization for hyper-parameter selection was implemented within DeSurv. The DeSurv R package is available at https://github.com/rashidlab/DeSurv (archived at Zenodo, DOI `10.5281/zenodo.20150929`). Reproducibility code, processed data, and the exact session-info file are available at https://github.com/rashidlab/DeSurv-paper (archived at Zenodo, DOI `10.5281/zenodo.20150946`); this includes `code/11_treated_cohort_analysis.R`, which generates the treated-cohort summary statistics above.

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Inclusion and Diversity. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies) and our access was limited to the de-identified expression and survival data they provide; we did not stratify analyses by sex, race, or ethnicity because consistent demographic annotations were not available across all cohorts. Sex and other demographic variables that could modify the prognostic relevance of the gene programs identified here remain important directions for follow-up work as more comprehensively annotated cohorts become available.

Author Contributions

Conceptualization: A.M.Y., N.U.R. Methodology: A.M.Y., A.Y., D.L., N.U.R. Software: A.M.Y. Formal analysis: A.M.Y. Investigation: A.M.Y. Data curation: A.M.Y., X.L.P. Visualization: A.M.Y. Writing – original draft: A.M.Y., N.U.R. Writing – review and editing: A.M.Y., A.Y., X.L.P., J.J.Y., D.L., N.U.R. Resources (PDAC domain expertise and biological interpretation): X.L.P., J.J.Y. Supervision: N.U.R. Funding acquisition: A.M.Y., N.U.R. All authors reviewed and approved the final manuscript.

Competing Interests

N.U.R. and J.J.Y. are named as inventors on US patents US20220127676A1 and US20250066856A1, which relate to pancreatic cancer subtype classification. The PurIST and DeCAF classifiers were used only as published, independently derived comparators: their subtype calls were obtained with the released classifiers and included as fixed covariate annotations, and the adjustment analyses using these calls are objective and reproducible from the deposited code. They are not components of DeSurv. The other authors declare no competing interests.

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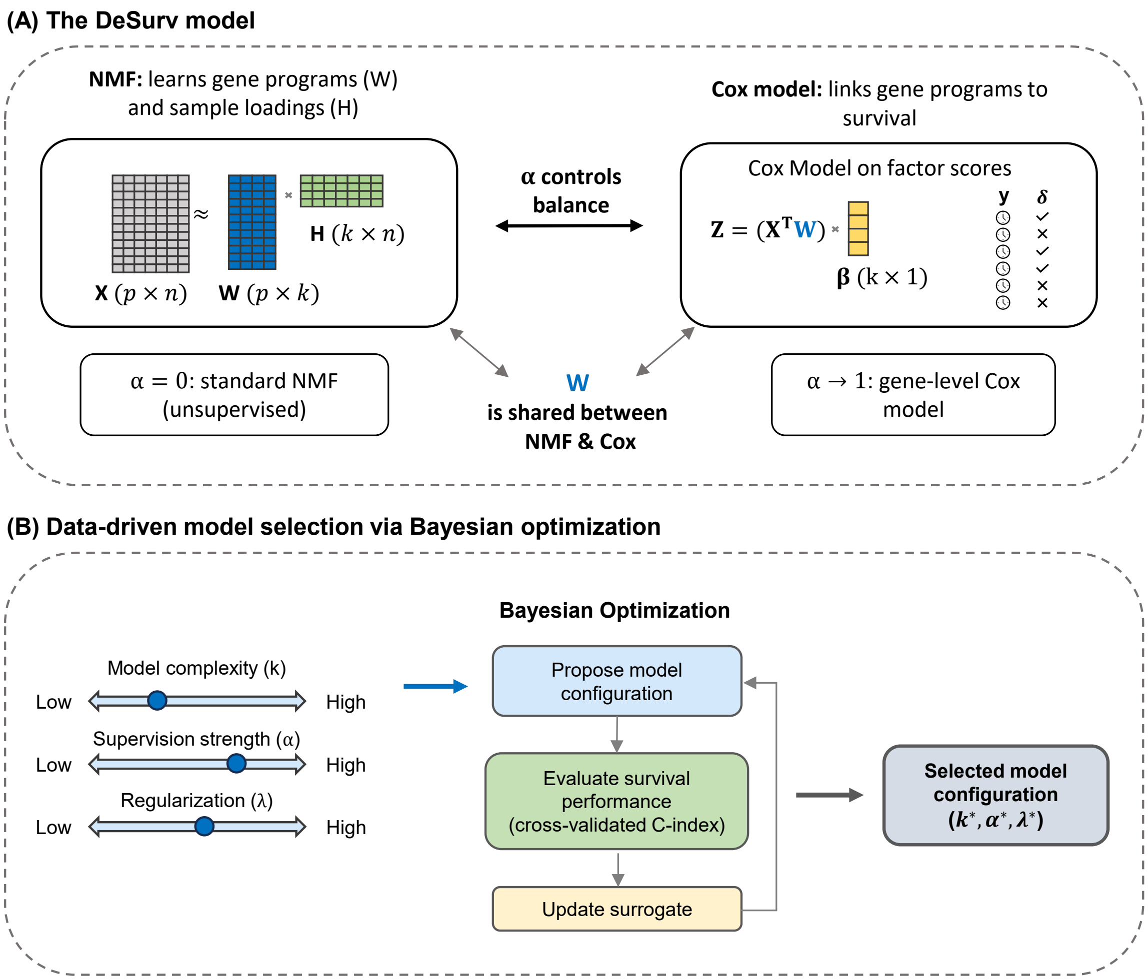

Figure 1: Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index.

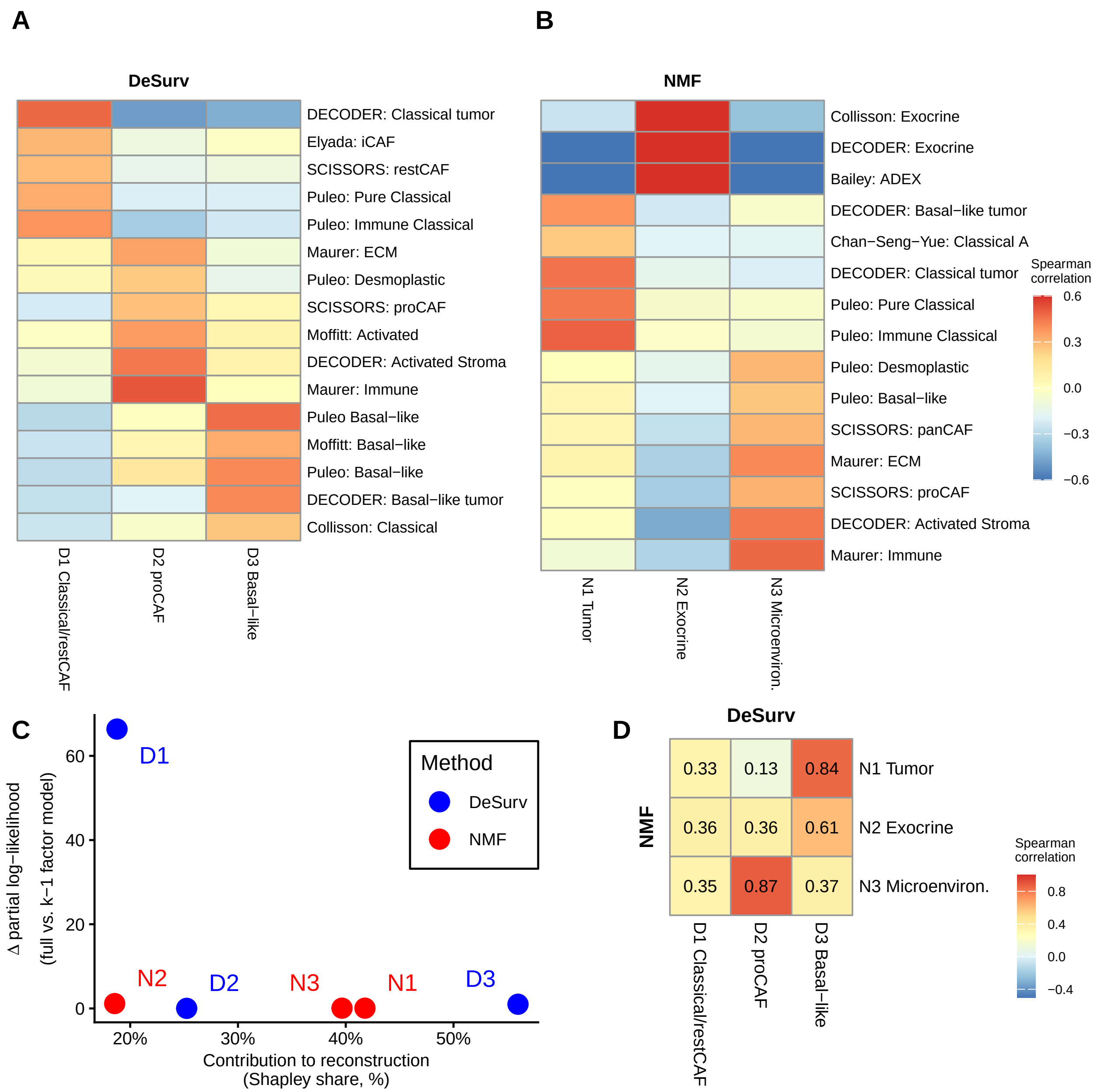

Figure 2: Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (A,B) Spearman correlations between factor gene rankings (top 50 genes per factor) and established PDAC gene programs for DeSurv (A) and standard NMF (B) at $k = 3$; only $|r| > 0.2$ shown. (C) Per-factor reconstruction contribution (normalized Shapley share, summing to 100%) versus survival contribution (Δ Cox partial log-likelihood on factor removal). (D) Pairwise correspondence between NMF and DeSurv factors (Spearman correlation of W-matrix gene loadings over all genes).

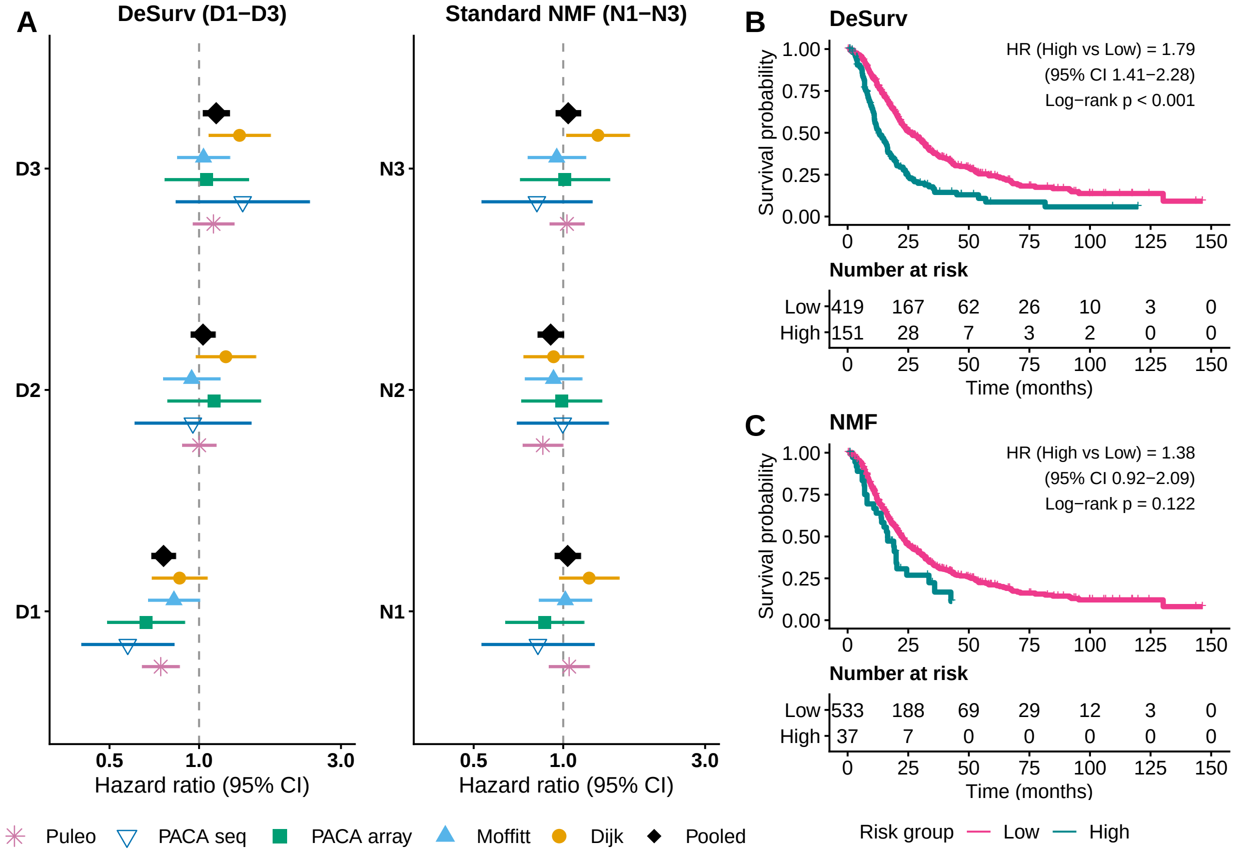

Figure 3: Survival-aligned programs generalize across independent PDAC cohorts (pooled $n = 616$ patients across 5 cohorts: Dijk $n = 90$; Moffitt GEO array $n = 123$; PACA-AU array $n = 63$; PACA-AU RNA-seq $n = 52$; Puleo array $n = 288$). (A) Forest plot of per-cohort univariate hazard ratios (95% CIs) for each factor; black diamonds denote inverse-variance-weighted pooled estimates. (B,C) Kaplan–Meier curves for pooled validation samples stratified by the training-derived multivariate linear predictor, dichotomized at a cross-validated optimal cutpoint: DeSurv (B) and standard NMF (C). No validation-cohort survival outcomes were used in training, scoring or hyperparameter selection.

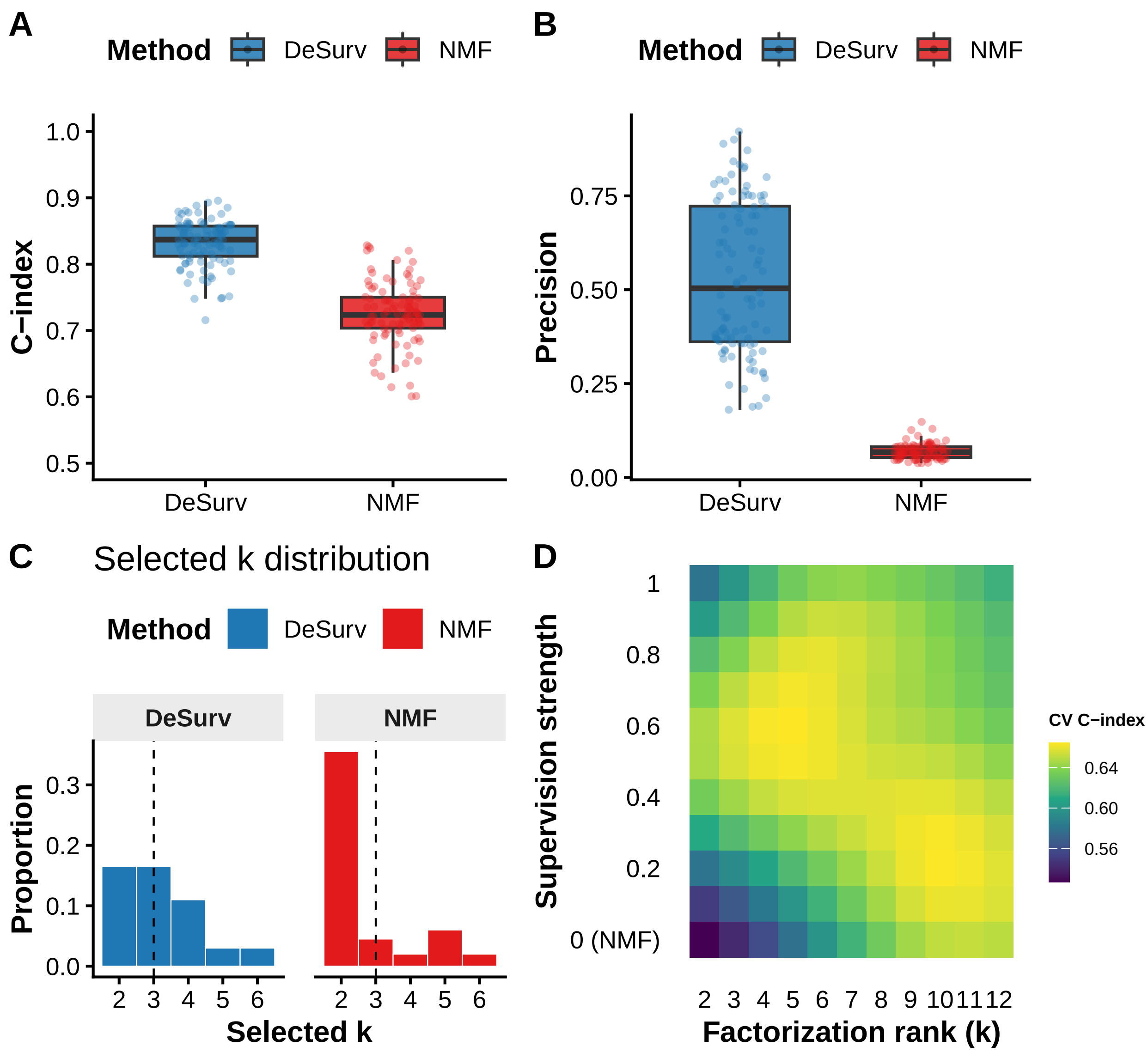

Figure 4: Survival supervision improves recovery of prognostic gene programs in simulations and PDAC training data. (A) Test-set concordance across 100 simulation replicates comparing DeSurv with standard NMF ($\alpha = 0$; $p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; paired Wilcoxon $P < 0.001$). (B) Precision of recovered prognostic gene programs (overlap between each true program and the top-weighted genes of its best-matching learned factor). In (A) and (B), boxes show median and interquartile range; whiskers extend to 1.5×IQR; points are individual replicates. (C) Distribution of selected ranks across replicates for DeSurv versus standard NMF. (D) Gaussian-process–predicted mean cross-validated C-index over factorization rank and supervision strength in PDAC training data (TCGA + CPTAC); this panel shows empirical tuning, not a simulation result.